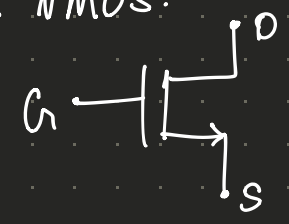
AMC

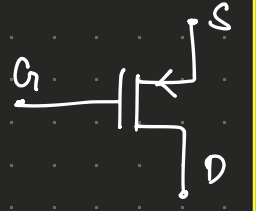
Unit - 2

Mosfets

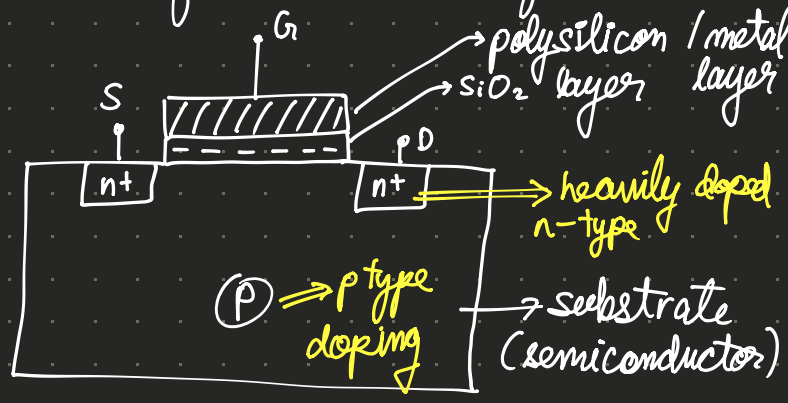
- NMOS:



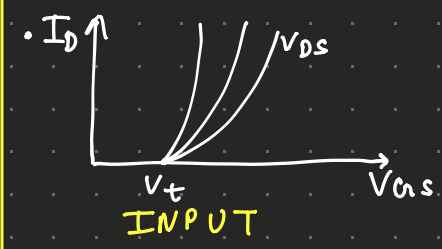
- PMOS

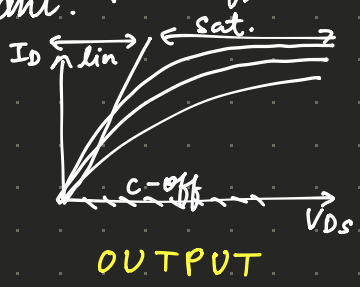


☆ Structure of N-channel Mosfet:



- for NMOS, gate has +ve voltage.
- We apply 2 voltages: → $V_{GS} : V_G - V_S$; → $V_{DS} : V_D - V_S$
- Modes of operation:
  1. $\boxed{\text{Cutoff Region: } V_{GS} < V_t}$ no channel formed
  2. $\boxed{\text{Linear/Triode: } V_{GS} > V_t \text{ ; } V_{DS} < V_{GS} - V_t \text{ /non saturation /resistive/ohmic region}}$
     current flows from drain to source.
  3. $\boxed{\text{Saturation Region: } V_{GS} > V_t \text{ ; } V_{DS} \geq V_{GS} - V_t}$
     channel width starts to decrease near the drain terminal → pinch off.
     I_D becomes constant.

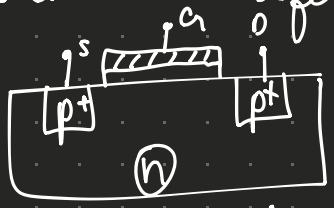




- Applications:

Region	Acts as
cut-off	Open Switch
linear	resistor
saturation	const. current source

☆ p channel mosfet

- 

- for pmos, gate voltage is −ve.
- Modes of Operation
  1. $\boxed{\text{Cutoff region: } V_{GS} > V_t}$
  2. $\boxed{\text{Linear region: } V_{GS} < V_t \text{ ; } V_{DS} > (V_{GS} - V_t)}$
     current flows from Source to Drain
  3. $\boxed{\text{Saturation Region: } V_{GS} < V_t \text{ ; } V_{DS} \leq (V_{GS} - V_t)}$

- V_{ov}: Overdrive Voltage.
  $V_{GS} - V_t = V_{ov}$.
- V_t: threshold voltage (min voltage applied to gate to form channel.
- V_T: thermal voltage

channel length: distance btw source & drain terminal

☆ Drain Current

- Linear Region: $I_D = K_n \left[(V_{GS} - V_t) V_{DS} - \frac{V_{DS}^2}{2} \right]$
- Saturation Region: $I_D = \frac{K_n}{2} [V_{GS} - V_t]^2$

$K_n = \mu_n C_{ox} \left(\frac{W}{L} \right)$ → aspect ratio

μ_n → mobility of charge; C_{ox} → oxide capacitance

$K_n' = \mu_n C_{ox} \ (A/V^2)$

- Mobility of Nmos > Mobility of Pmos
  $\mu_n > \mu_p$.

- $C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} = \frac{3.9\,\varepsilon_0}{t_{ox}}$ → oxide thickness

$\varepsilon_0 : 8.854 \times 10^{-12}$ F/m

- $dQ = C_{ox} \cdot w \cdot dx\,[V_{GS} - V_t - V(x)]$

applied voltage at the strip.

★ Channel length modulation (CLM)

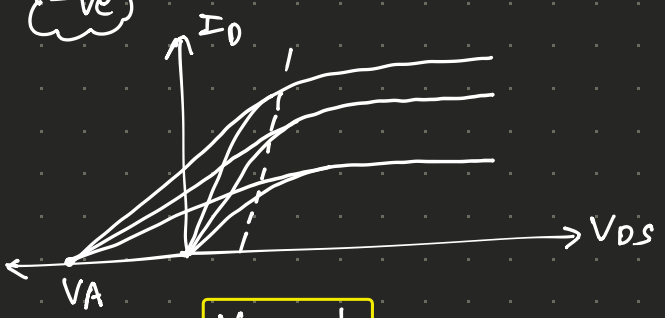
- as $V_d \uparrow$, channel length $\downarrow$ ⟹

- process dependent parameter → $\boxed{\frac{\lambda'}{L} = \lambda}$

- $$\boxed{I_D = \frac{K_n}{2}(V_{GS} - V_t)^2(1 + \lambda V_{DS})}$$

- Early Voltage (V_A): drain to source voltage where $I_D = 0$ when transistor's in saturation.

(−ve)



$$\boxed{V_A = \frac{1}{\lambda}}$$

- Transconductance (g_m)

$$\boxed{g_m = \frac{I_D}{V_{GS}}}$$ $\frac{\text{o/p current}}{\text{i/p voltage}}$

* defined in the saturation region *

$$\boxed{g_m = K_n'\left(\frac{W}{L}\right) V_{ov}}$$

$$\boxed{g_m = \frac{2 I_D}{V_{ov}}}$$

$$\boxed{g_m = \sqrt{2 K_n' \left(\frac{W}{L}\right) I_D}}$$

★ Resistance offered by the transistor

1. In ohmic region: r_{ds}

$$\boxed{r_{ds} = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{K_n (V_{GS} - V_t)}}$$

2. In saturation region: r_o → resistance offered when there is CLM.

$$\boxed{r_o = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\lambda I_D} = \frac{V_A}{I_D}}$$

- Body effect: variation of V_t due to variation of bulk or substrate voltage.

$$\boxed{V_t = V_{t0} + \gamma\left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f}\right]}$$

threshold voltage (new) — threshold voltage (old) — body effect parameter — source voltage − bulk voltage — fermi voltage

if $V_{SB} = 0$; $V_t = V_{t0}$